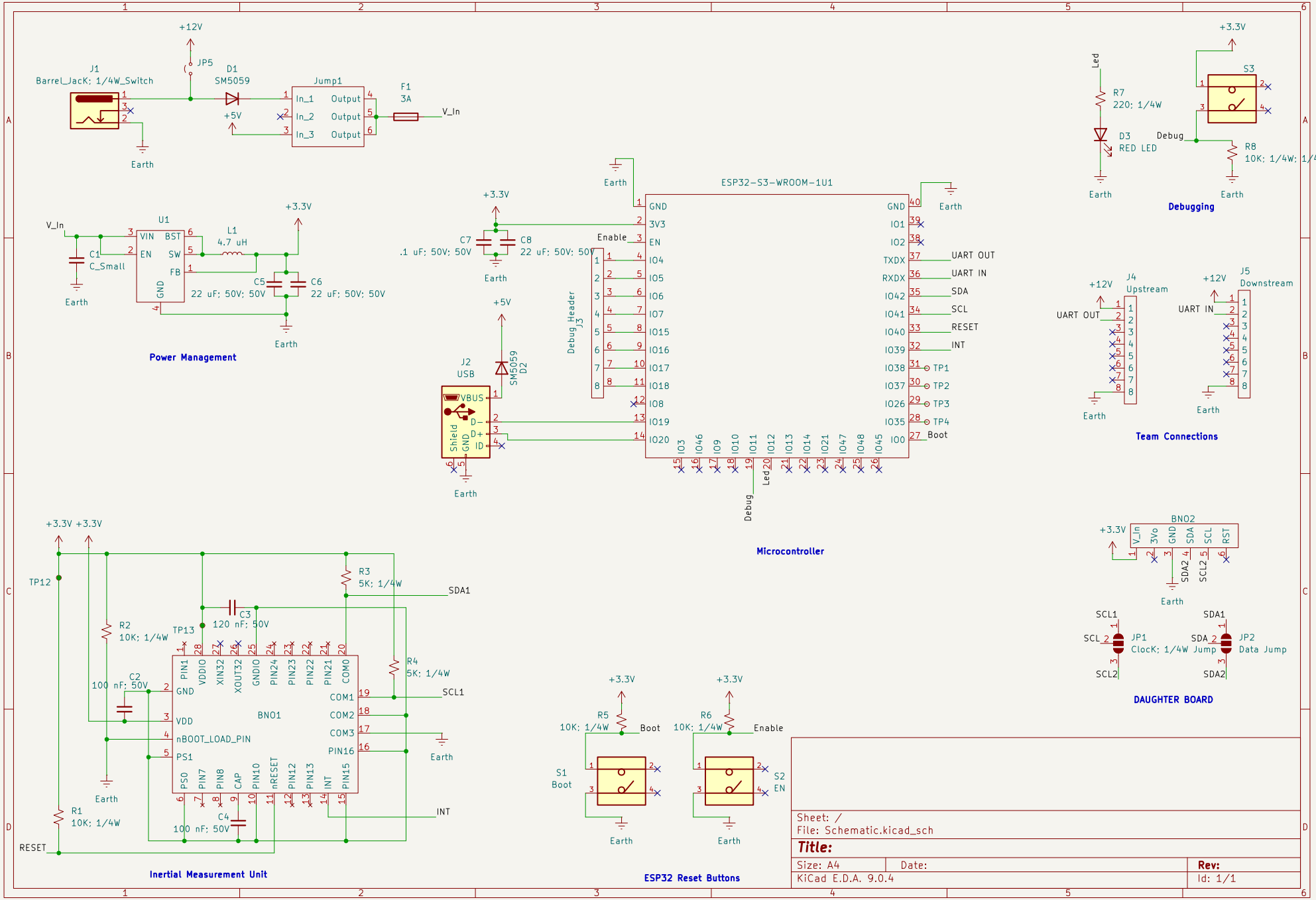
+12V
J1
Barrel_JacK; 1/4W_Switch
JP5
D1
SM5059
+5V
Jump1
In_1
In_2
In_3
Output
Output
Output
F1
3A
V_In
Earth
U1
VIN
BST
EN
SW
FB
GND
L1
4.7 uH
+3.3V
C1
C_Small
C5
22 uF; 50V; 50V
C6
22 uF; 50V; 50V
Power Management
ESP32-S3-WROOM-1U1
GND
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19
IO20
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
GND
IO1
IO2
TXDX
RXDX
IO42
IO41
IO40
IO39
IO38
IO37
IO26
IO35
IO0
UART OUT
UART IN
SDA
SCL
RESET
INT
TP1
TP2
TP3
TP4
Boot
Enable
Led
Debug
C7
.1 uF; 50V; 50V
C8
22 uF; 50V; 50V
Debug_Header
J3
J2
USB
VBUS
D-
D+
ID
Shield
GND
SM5059
D2
Microcontroller
Led
R7
220; 1/4W
D3
RED LED
S3
Debug
R8
10K; 1/4W; 1/4W
Debugging
J4
Upstream
J5
Downstream
UART OUT
UART IN
Team Connections
BNO2
V_In
3Vo
GND
SDA
SCL
RST
SDA2
SCL2
JP1
ClocK; 1/4W Jump
JP2
Data Jump
DAUGHTER BOARD
TP12
R2
10K; 1/4W
C2
100 nF; 50V
TP13
C3
120 nF; 50V
R3
5K; 1/4W
R4
5K; 1/4W
SDA1
SCL1
INT
RESET
R1
10K; 1/4W
C4
100 nF; 50V
BNO1
PIN1
VDDIO
XIN32
XOUT32
GNDIO
PIN24
PIN23
PIN22
PIN21
COM0
COM1
COM2
COM3
PIN16
GND
VDD
nBOOT_LOAD_PIN
PS1
PS0
PIN7
PIN8
CAP
PIN10
nRESET
PIN12
PIN13
INT
PIN15
Inertial Measurement Unit
R5
10K; 1/4W
Boot
R6
10K; 1/4W
Enable
S1
Boot
S2
EN
ESP32 Reset Buttons
Sheet: /
File: Schematic.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.4
Id: 1/1